

ENEL700 Tutorial 10

1. What is channel coding? Identify its two main functions.

>>Solution, refer to lecturer slide 323. Channel encoding refers to the process of error detection and correction involves adding extra bits to the data characters to be transmitted. The two main functions are error detection and error correction.

2. What are the two types of error correction codes? Briefly discuss their differences.

>>Solution, refer to lecture slides 327-333, block codes and convolutional codes are the two types of error correction codes.

For block code, a block of k data bits is encoded by a code word of n bits ($n > k$). For each k sequence of bits there is an unique n bit word.

For convolutional code, data bits are continuously generated. The output bits not only depend on the present bit but also on few earlier bits.

3. Discuss the main difference between multiplexing and duplexing.

>>Solution, refer to lecture slides 334-365, multiplexing is the process of simultaneously transmitting two or more individual signals over a single communication channel. While duplexing is the method by which two-way communications are handled.

4. Can we apply frequency division multiplexing in a lecture theatre? If so, discuss how it works.

>>Solution, yes, different people have different vocal frequency range. For example, the voiced speech of a typical adult male will have a fundamental frequency from 85 to 180 Hz, and that of a typical adult female from 165 to 255 Hz. In this way, multiple pairs of conversations can be allowed if they are using different vocal frequency.